

Services Consultant Architect Practice Lab 3

IMPORTANT: The **approximate build time for this lab is 10 minutes**. Please be patient. If the lab does not build after 20 minutes, try refreshing the page and rebuilding it again. If it does not build on the second attempt, contact ellearn@splunk.com.

Lab Overview

Maximum Estimated Duration – 4 Hours

This lab will focus on the configuration of data forwarding through an intermediate tier of universal forwarders and routing data to specific indexes based on the contents of the raw events.

You will configure a single Search Head, two Indexers and a Monitoring Console as in the previous lab. There are four servers that will need to be configured to forward locally generated data through two load balanced intermediate universal forwarders to your indexers.

Your search head will be configured as the deployment server to manage the configuration and installation of apps on all universal forwarders.

Data is being generated on each External Forwarder in the following folder: `/opt/data/`

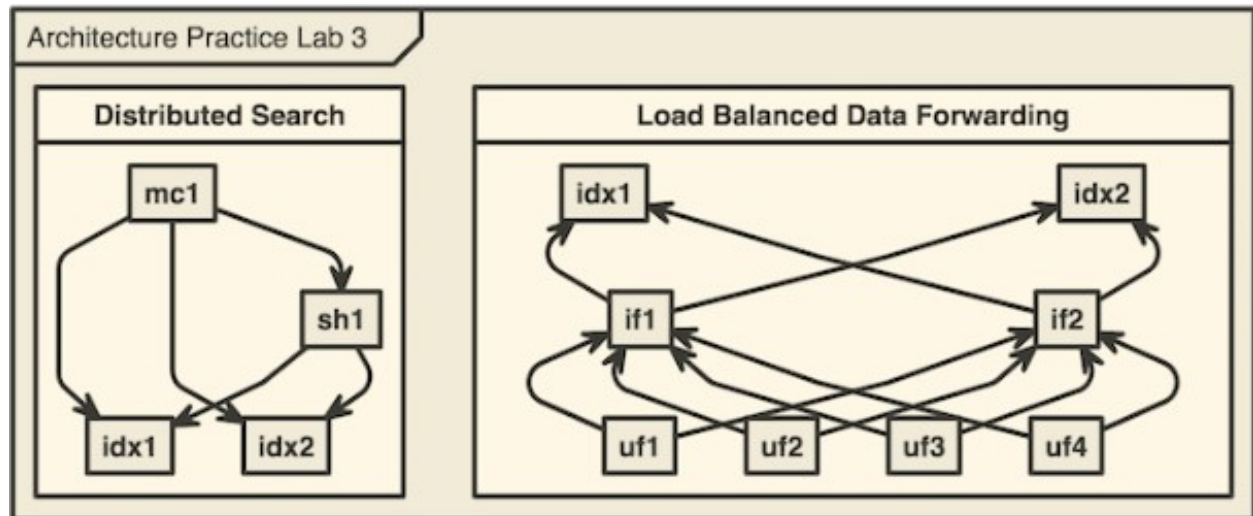
The data will need to be forwarded to different indexes based on the following rules:

Index	Rule
Novels	default destination
Sherlock	event contains the text "Sherlock"
Watson	event contains the text "Watson"
Moriarty	event contains the text "Moriarty"

Host Name	Role
Ap3-xxxx-sh1	Search Head (Deployment Server)
Ap3-xxxx-mc1	Monitoring Console
Ap3-xxxx-idx1	Indexer
Ap3-xxxx-idx2	Indexer
Ap3-xxxx-uf1	External Universal Forwarder
Ap3-xxxx-uf2	External Universal Forwarder
Ap3-xxxx-uf3	External Universal Forwarder
Ap3-xxxx-uf4	External Universal Forwarder

Ap3-xxxx-if1
Ap3-xxxx-if2

Internal Intermediate Universal Forwarder
Internal Intermediate Universal Forwarder



Configure your Distributed Search Infrastructure (sh1, idx1, idx2, mc1)

Use the same steps from the Services Consultant Architecture Practice Lab 1 to configure your infrastructure.

On each host – excluding your forwarder hosts:

1. Create a splunk user.
2. Unarchive Splunk Enterprise package in /opt.
3. Install Splunk as Root.
4. Change the default admin password. (Use the password provided in your Server list)
5. Run Splunk as a non-root User.
6. Enable Splunk to start on boot as the splunk user.
7. Configure your indexers.
8. Configure distributed search.
9. Configure the Monitoring Console.

Deploy the Universal Forwarders

You now have 6 instances to install the universal forwarder on. (uf1, uf2, uf3, uf4, if1, if2).

Perform the following:

1. Create a splunk user.
2. Unarchive Splunk Universal Forwarder package in /opt.
3. Install Splunk as Root
4. Change the default admin password. (Use the password provided in your Server list)
5. Ensure Splunk will run as the splunk user.
6. Use the org_all_deployentclient template to create and configure an ap3_deploymentclient app that calls into the Search Head based deployment server.

The following servers will also need to be configured as deployment clients:

- Monitoring Console
- Indexers
- Universal Forwarders

7. Using the Forwarder Management UI on the Search Head confirm that all 9 clients have phoned home to the Deployment Server.

ServerClass 1 – Internal Forwarders

1. Use the base configuration template "org_all_forwarder_outputs" to configure your Intermediate Forwarders to send their logs to your Indexers.
The app should be renamed ap3_internal_forwarder_outputs.
2. Update the server list under the [tcpout:primary_indexers] stanza in outputs.conf with your indexers.
3. Copy the same forwarder outputs app to both the apps and deployment-apps folders on your search head.

This will ensure that the Search Head internal logs are forwarded to the indexing tier in the same manner as the forwarder.

4. Restart Splunk to ensure you configuration app is applied on the Search Head.

In this section you will use the Forwarder Management UI to configure your internal instances to send their logs to the indexing tier.

5. Create a new Server Class titled "Internal_Forwarders".
6. Add a whitelist of "*if*", "*mc1*".
7. Add your updated ap3_internal_forwarder_outputs app to the Internal_Forwarders server class.
8. Configure the server class to restart splunkd on update.

9. From your Search Head run the following search to check your work:
`index=_internal | stats count by host`

You should see results from all 6 instances.

10. Use the base configuration template "org_APP_TEMPLATE" to configure your Intermediate forwarders to send data without a throughput limit.
11. Rename the template app to `ap3_intermediate_forwarder_limits`.
12. Add a local/limits.conf file and add a stanza with the title "[thruput]" and a setting of:
`"maxKBps = 0"`.
13. Use the base configuration template "org_APP_TEMPLATE" to configure your Intermediate forwarders to send data using multiple ingestion pipelines.

To ensure even data spread we should always have 2x the number of pipelines compared to indexers. As we have 2 indexers and 2 forwarders, our forwarders need to be configured with 2 ingestion pipelines.

14. The template app should be renamed `"ap3_all_intermediate_forwarder_pipelines"`.
15. The app should contain a server.conf file with the following: [general]
`parallelIngestionPipelines = 2`
16. Use the base configuration template "org_APP_TEMPLATE" to configure your Intermediate forwarders to receive data from the primary forwarders.
17. Rename the template app to `ap3_intermediate_forwarder_inputs`.
18. Add a local/inputs.conf file and add a stanza with the following value: `[splunktcp://9997]`.

Now configure the apps to be deployed

1. Create a new server class titled "Intermediate_Forwarders".
2. Add a whitelist of: `"*if*"`.
3. Add the following new apps to the server class:
4. `ap3_intermediate_forwarder_limits`
5. `ap3_all_intermediate_forwarder_pipelines`
6. `ap3_intermediate_forwarder_inputs`
7. Configure the server class to restart splunkd on update.

ServerClass 2 – External Forwarders

1. Use the base configuration template "org_all_forwarder_outputs" to configure your Universal Forwarders to send their logs to your Intermediate Forwarders.

2. Rename the template app to `ap3_external_forwarder_outputs`.
3. Set the `defaultGroup` key to a value of `intermediate_forwarders` in `outputs.conf`. Change the `[tcpout:primary_indexers]` stanza to `[tcpout:intermediate_forwarders]`.
4. Update the server list under the `[tcpout:intermediate_forwarders]` stanza in `outputs.conf` with your intermediate forwarders.
5. Using the Forwarder Management UI configure your external forwarders with the app you've just created by configuring a serverclass to deploy your Remote Forwarder apps.
6. Create a new serverclass titled "Remote_Forwarders".
7. Add a whitelist of `"*uf"`.
8. Add your all `ap3_external_forwarder_outputs` app to the serverclass.
9. Configure the serverclass to restart `splunkd` on update.
10. From your search head run the following search to check your work.
`index=_internal | stats count by host`

You should see results from all 10 of your instances.

Index & Volume Configuration

1. Use the base configuration app template `"org_indexer_volume_indexes"` to configure your indexers to store the indexed data in a non-default location outside of the splunk folder:
2. Rename the app to `ap3_indexer_volume_indexes`
3. Edit the path for the primary and secondary volumes in the `local/indexes.conf` setting it to `/opt/splunkdata`.
4. On both of your indexers create the `/opt/splunkdata` folder and set the permissions on it to be owned by your splunk user.
5. Using the Forwarder Management UI configure your indexers with the `ap3_indexer_volume_indexes` app you've just created by configuring a serverclass to deploy it.
6. Create a new serverclass titled "All_Indexers"
7. Add a whitelist of `"*idx"`
8. Add your `ap3_indexer_volume_indexes` app to the serverclass
9. Configure the serverclass to restart `splunkd` on update
10. Create a copy of the `ap3_indexer_volume_indexes` app and name it `ap3_search_volume_indexes`.

Normally the search version of this app would point to different storage volumes or locations than the indexers.

For this practice lab we will keep them the same.

11. On both your search head and monitoring console create the /opt/splunkdata folder and set the permissions on it to be owned by your splunk user.
12. Create a new serverclass titled "All_Search_Heads"
13. Add a whitelist of "*-mc*"
14. Add your all ap3_search_volume_indexes app to the serverclass
15. Configure the serverclass to restart splunkd on update
16. Add your all ap3_search_volume_indexes app to the apps folder of your search head/deployment server instance and restart splunkd
17. Use the base configuration app template "org_all_indexes" to configure your indexers with the required indexes.
18. Rename the app to ap3_all_indexes
19. Under the "# CUSTOMER INDEXES" comment in the indexes.conf file add the following stanza's:

```
[sherlock]
homePath = volume:primary/sherlock/db
coldPath = volume:primary/sherlock/colddb
thawedPath = $SPLUNK_DB/sherlock/thaweddb
```

```
[watson]
homePath = volume:primary/watson/db
coldPath = volume:primary/watson/colddb
thawedPath = $SPLUNK_DB/watson/thaweddb
```

```
[moriarty]
homePath = volume:primary/moriarty/db coldPath
= volume:primary/moriarty/colddb
thawedPath = $SPLUNK_DB/moriarty/thaweddb
```

```
[novels]
homePath = volume:primary/novels/db
coldPath = volume:primary/novels/colddb
thawedPath = $SPLUNK_DB/novels/thaweddb
```

20. Using the Forwarder Management UI configure your indexers and search heads with the ap3_all_indexes app you've just created by adding it to the appropriate serverclasses.
21. Add your all ap3_all_indexes app to the following serverclasses:
 - "All_Search_Heads"
 - "All_Indexers"

1. Use the base configuration app template "org_APP_TEMPLATE" to configure your indexers to route the data to the correct indexes based on the contents of the data.
2. The template app should be renamed ap3_sherlock_props
3. Create a local/transforms.conf and add the following:

```
[sherlock_routing]
REGEX = Sherlock
DEST_KEY = _MetaData:Index
FORMAT = sherlock
```

```
[watson_routing]
REGEX = Watson
DEST_KEY = _MetaData:Index
FORMAT = Watson
```

```
[moriarty_routing]
REGEX = Moriarty
DEST_KEY = _MetaData:Index
FORMAT = Moriarty
```

4. Create a local/props.conf and add the following:


```
[sherlock]
TRANSFORMS-routing=sherlock_routing, watson_routing, moriarty_routing
```
5. Add the ap3_sherlock_props app to your "All_Indexers" serverclass.
6. Use the base configuration app template "org_dept_app_inputs" to configure your forwarders to monitor the input files.
7. The template app should be renamed ap3_sherlock_inputs
8. Replace the local/inputs.conf and add a stanza with the following:


```
[monitor:///opt/data/*.log] index=novels sourcetype=sherlock
```
9. Add the ap3_sherlock_inputs app to your Remote_Forwarders serverclass.
10. From your search Head run the following search:


```
index=* | stats count by index
```

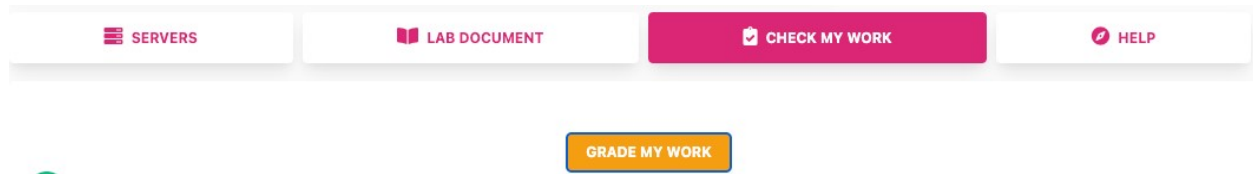
You should see results for all four new indexes.

The following search will allow you to check data is coming from each external forwarder:

```
index=* | stats count by host
```

Grading Your Work

Grading of the lab is done through the same interface where your student instances are displayed.



Grading of the labs is an intensive process involving many calls to the instances. This may take several minutes to complete. Please be patient. If the grading does not complete in 5 minutes, please reach out to ellearn@splunk.com for assistance.

For best results, make sure you complete all of the lab steps before grading the lab.

1. Click the Check My Work Tab.
2. Click the Grade My Work button.

After a few minutes, your results will be displayed.

You must pass all graded items to receive credit for the lab. Resolve any failed grading items and re-issue the grading check.

If you pass all of the lab grading checks, the Learning Management System (LMS) will terminate the browser window. Log back into the LMS to view your completed status for the lab.